TERRA AI — EcoPoints Architecture & Gemini Integration

How the Intelligent Vision & Category Scoring Engine Identifies Actions and Awards Points

1. System Overview: What Happens Without a Caption

In a conventional platform, users must manually select a category from a dropdown and draft an explanatory caption. If a user uploads an image with **no caption**, conventional systems default to generic fallback categories (such as 'other') with baseline minimum rewards.

With Gemini 3.5 Multimodal Vision integrated into TERRA AI, caption text is completely optional. When a photo or video is uploaded, the raw media buffer is evaluated directly by Gemini to autonomously classify the activity, detect authentic environmental stewardship, generate an informative narrative, and compute the corresponding EcoPoints.

2. Gemini Vision Classification & Points Matrix

Detected Scene (Gemini Vision)	Assigned Category	Impact Tier	Points Awarded
Trash bags, plastic bottles collected on beach/lake/park	waste_cleanup	Very High (Immediate pollution removal)	**+35 to +50 pts**
Fresh sapling planted in soil, root balling, tree digging	tree_planting	High (Long-term CO2 sequestration)	**+25 to +40 pts**
Coconut shell crafts, upcycled bottle planters, candle cups	upcycling	Medium-High (Circular waste diversion)	**+25 to +30 pts**
Balcony composting, earthworms, kitchen scrap bin, veggie bed	organic_gardening	Medium (Methane reduction & soil health)	**+20 to +25 pts**
Segregated paper, metal, e-waste boxes sent to recyclers	recycling	Standard (Closed-loop lifecycle)	**+15 to +20 pts**
Bicycle transit, bus ticket, reusable bottle, energy switch	sustainable_transport / other	Daily Habit (Carbon reduction)	**+10 to +15 pts**
Monthly emissions log completed in Carbon Calculator	carbon_log (System)	Periodic Tracking	**+30 pts**
New citizen registration on platform	onboarding (System)	Welcome Bonus	**+50 pts**

3. AI Anti-Fraud & Authenticity Verification

To prevent abuse (such as uploading irrelevant screenshots or random selfie photos to farm points), Gemini evaluates image authenticity before committing rewards to the database:

• **Relevance Verification:** Non-eco photos (coffee mugs, cars, computer monitors) return isEcoAction: false and 0 points with a polite prompt.

• **Scale-Aware Multiplier:** Gemini measures volume and scope (e.g. 10 bags of litter collected by a team vs. 1 picked up plastic wrapper), adjusting points between 35 and 50 points proportionally.

• **Automated Storytelling:** Even if the caption field was left blank by the citizen, Gemini auto-generates an inspiring 1-sentence caption to display on the community GreenFeed.

4. Stewardship Progression & Badge Milestones

Stewardship Level	Points Required	Earnable Achievement Badge
Level 1: Green Seedling	0 – 99 pts	■ **Green Seedling (Welcome)**
Level 2: Eco Warrior	100 – 249 pts	■■ **Eco Warrior (50 pts)**
Level 3: Eco Champion	250 – 499 pts	■■ **Clean Champion (100 pts)**
Level 4: Earth Protector	500 – 749 pts	■ **Carbon Cutter (150 pts)**
Level 5: Planet Guardian	750+ pts	■ **Circular Hero (200 pts)**